

Phase 1 - Creating the Project

Time	Start date	Finish date	Status
1 week			Not started

Goal: Create the real project structure so that the backend, frontend and database all run on your laptop.

Real-life example: Before building a house you prepare the land, bring the tools and draw the plan. This phase is exactly that.

Step checklist

#	Step	Done
1	Install JDK 21, IntelliJ IDEA, VS Code, Node.js (LTS), Docker Desktop, Postman, Git	☐
2	Create the GitHub repository "eventhub" with backend/ and frontend/ folders	☐
3	Generate the Spring Boot project from start.spring.io (Web, Data JPA, Validation, Security, MySQL, Flyway, Actuator, Mail, Cache, Lombok)	☐
4	Create the React app with Vite; install Tailwind CSS, React Router, Redux Toolkit, Axios	☐
5	Write docker-compose.yml with MySQL 8 and Mailpit (fake email inbox)	☐
6	Connect Spring Boot to MySQL in application.yml; create dev and prod profiles	☐
7	Add a /api/health endpoint and Swagger UI	☐
8	Add GitHub Actions: build and test automatically on every push	☐
9	Draw the page sketches and database diagram on paper (keep a photo in the repo)	☐

Output of this phase

- Running Spring Boot app at localhost:8080
- Running React app at localhost:5173
- MySQL and Mailpit running in Docker
- Green CI tick on GitHub

Done when (check before moving on)

Check	Passed
The React page shows "Backend: UP"	☐
docker compose up starts the database with one command	☐

Skills you practise

Project setup, Maven, npm, Docker, configuration, CI/CD

Where your work goes

backend\ and frontend\ (project skeleton), docker-compose.yml, .github\workflows\

My notes
